Momentum Strategies

About You

- What’s your name?
- Do you already trade algorithmically?
- Do you trade stocks, futures, or currencies?

What Cause Momentum?

- Futures: Persistence of roll returns.
- Slow diffusion, analysis and acceptance of new information. E.g. PEAD
- Forced sales or purchases of assets by index/mutual/hedge/exchange-traded funds.
  - Contagion due to
    - Risk management
    - Investors redemption/subscription
  - Index composition changes
  - Levered ETFs: Forced rebalancing at market close.

HFT and “Momentum Ignition”

- HFT: Market manipulation ignites momentum.
  - Quote matching
  - Flipping
  - Stop hunting
  - Front-running order flow
- We will examine trading strategies exploiting each cause of momentum.

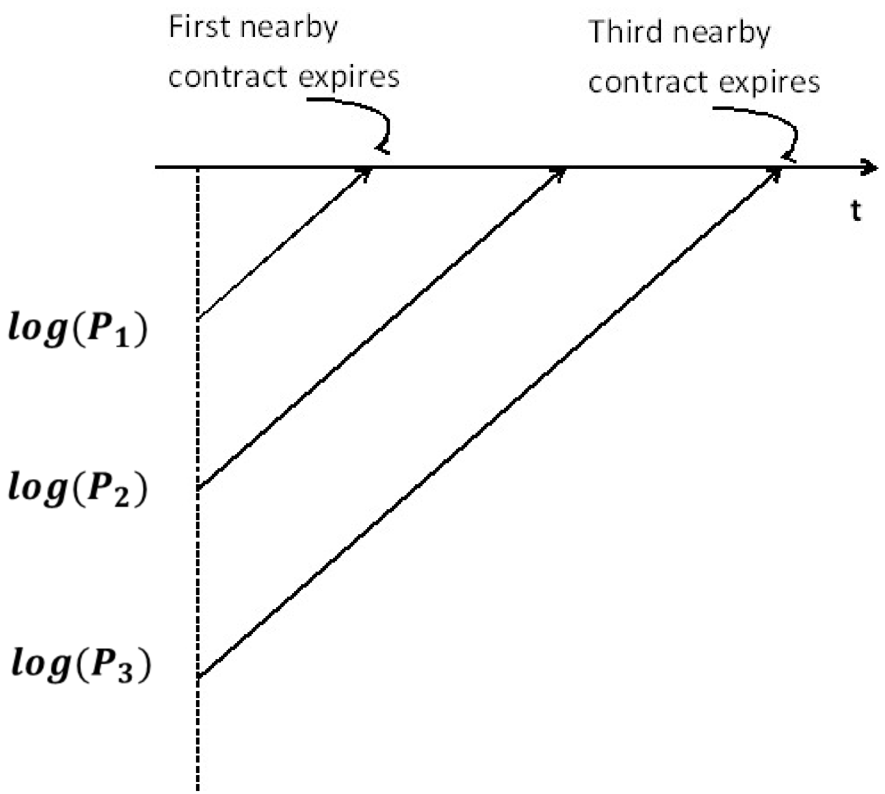
Why Python?

- Python is free!
- A simple programming language
- Pandas extension: library built for data reading and analysis.
- Can be compiled to C/C++ for fast execution.

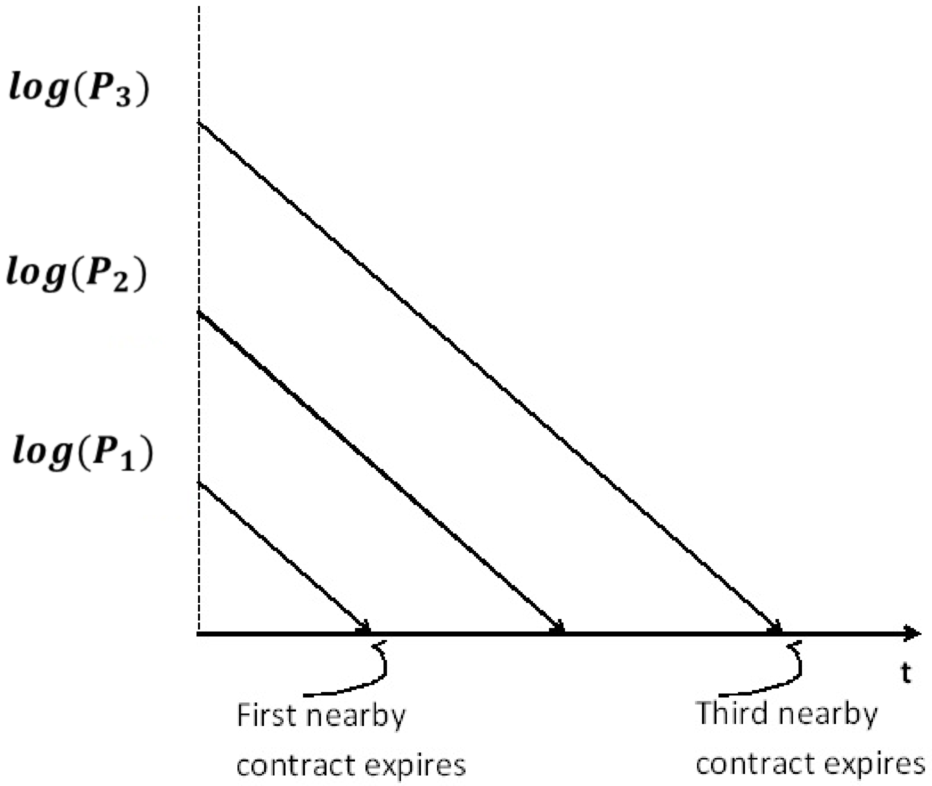
Roll Returns As Driver of Futures Momentum

- Total returns in futures = returns of spot price+roll returns.
  - Even if the spot price is unchanged, the futures price can still change.
  - If roll return is positive: (normal) backwardation.
  - If roll return is negative: contango.

Backwardation



Contango



$log(P_3)$
$log(P_2)$
$log(P_1)$
t
First nearby contract expires
Third nearby contract expires

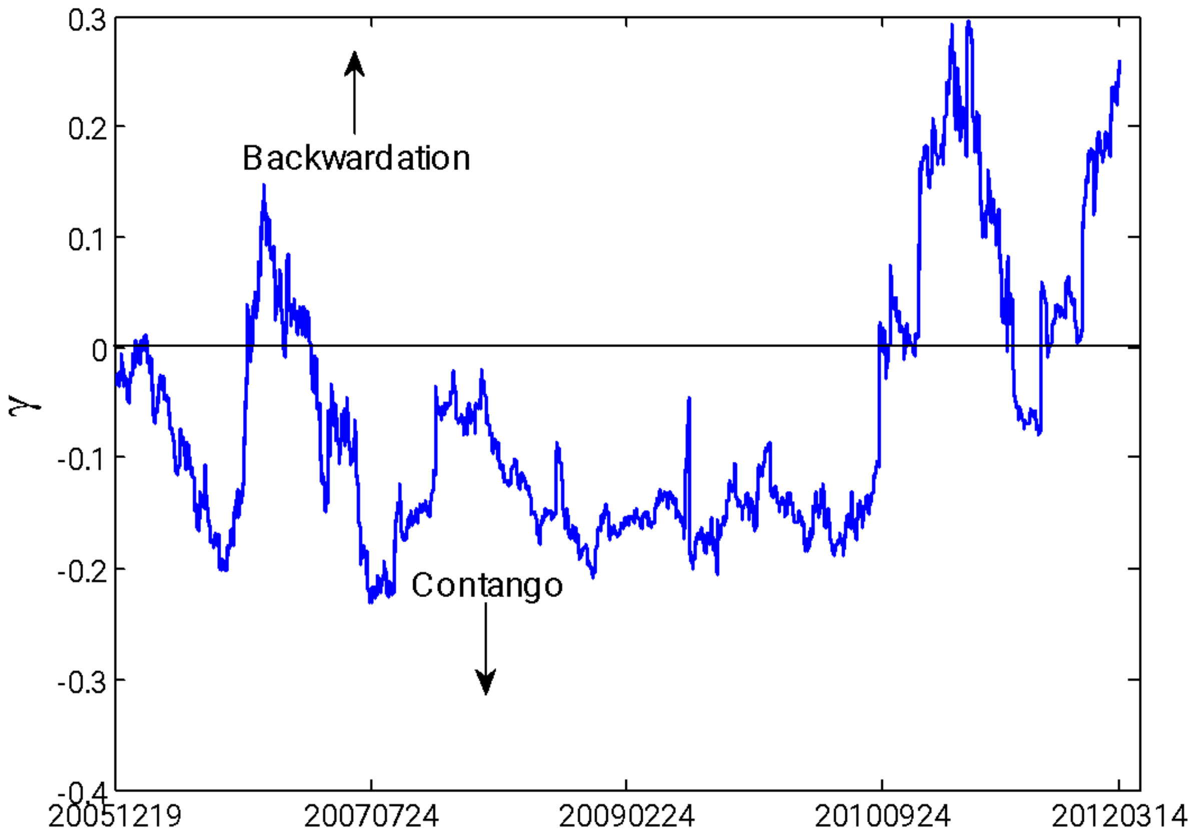
Forward Curves

- "Forward curve" of futures = plots of futures prices against time-to-maturity at a fixed time in history.
  - As opposed to our plots of prices evolving in time.
- Also called the "term structure" of futures.
- *Exercise:* Sketch the forward curves of futures in backwardation and contango, respectively.

- Assume both spot and roll returns are constant over time
- $F(t, T) = c\ e^{at} \exp(y(t-T))$
- Where t is current date
- T is expiration date
- C, a(spot return), and y(roll return) are constants

- Example corn futures

Roll returns as a function of time



Spot vs Roll Returns

- α =slope of log cash prices vs time

  =2.8%
- γ =slope of forward curve (log futures prices vs time-to-maturity).
- Note during longer periods, $|\gamma| \gg \alpha$
  - Very favorable for momentum trading!

Types of Momentum

- Time series momentum:
  - Past returns of a price series are positively correlated with future returns.
  - E.g. A stock that went up will go higher.
- Cross-sectional momentum:
  - Past *relative* returns are positively correlated with future *relative* returns.
  - I.e. Past returns of an instrument that out(under)-performs another instrument will continue to do so.
  - E.g. NVDA outperformed AAPL last year, though both went up. NVDA will continue to outperform AAPL this year, even though we are in a bear market for tech stocks.

Roll Returns and Futures Momentum

- Roll returns are much less volatile than spot returns, and they maintain the same sign for long periods.
- If roll returns dominate total returns of a future (at long time-scale) $\Rightarrow$ time-series momentum.
- Even if spot returns dominate total returns, as long as they are not anti-correlated with roll returns, they can be arbitraged away in a long-short portfolio $\Rightarrow$ Cross-Sectional Momentum.

Extraction of Roll Returns

- Is there a more reliable way to extract roll returns?
  - Yes: arbitrage between future and spot prices.
    - E.g. GC (gold futures) vs GLD (gold trust ETF)
  - See "What's Behind the Metals Warehouse Debate?", Neeraj Batra, tabbforum.com, 4 Oct, 2013

  - *Exercise*:
    - GC is usually in contango: just short it while buy and hold GLD.
    - Find the annualized average return of this strategy.
    - Find the Sharpe ratio, assuming risk-free rate is 2% p.a.

Extraction of Roll Returns

- What if there is no (readily traded) underlying asset?
  - Can still arbitrage between future and another traded instrument highly (anti-)correlated with the spot return of the future.
  - *Exercise*: Can you think of examples for
    - Futures vs ETF?
    - Futures vs futures?
    - Futures vs X?
    - ETF vs ETF?

Exercise: XLE vs USO

- Buy XLE and short USO whenever CL is in contango, and vice versa.
  - Find the annualized average returns and Sharpe ratio.

The Case of VX

- VX has large roll return
  - 50% annualized
- VIX is highly anti-correlated with ES
  - Correlation between daily returns=-75%
- ES has little roll return of its own.
- Arbitrage between VX and ES should yield much of the roll return of VX!
- First, find the hedge ratio appropriate for this arbitrage

VX-ES

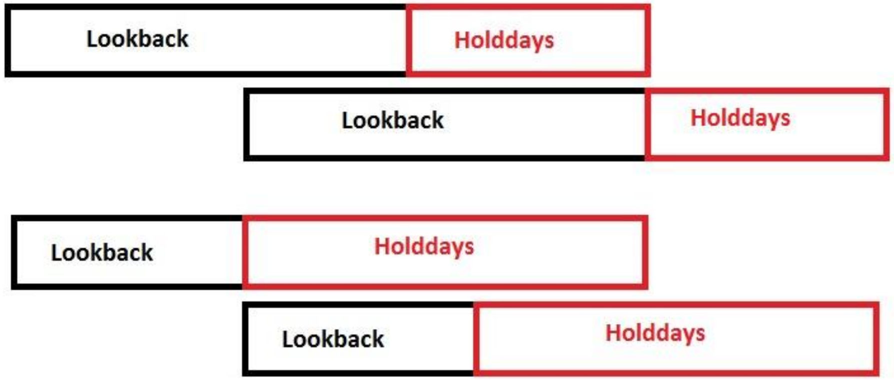
- Strategy:
  - If VX’s roll return > threshold: Buy VX, buy ES
  - If VX’s roll return < -threshold: Sell VX, sell ES
- *Exercise*: How should we compute roll returns for VX?

Other ways to extract the futures momentum

- Time series and cross-sectional momentum can still be extracted as long as long-term spot returns are uncorrelated or smaller in magnitude than roll returns.
- First, we need some tests to see if this is possible for a specific future.

Testing for Time-Series Momentum

- The literal way: Test for positive serial correlation of returns.
  - First, find non-overlapping periods between backward (lookback) and forward (holding days) returns.



TU Momentum

- Alternative tests: Hurst exponent and Variance Ratio Test.
- **Hurst exponent**: Test for long-term momentum
  - Compute the variance of the log prices $z(t)$
    $$\mathrm{Var}(\tau) = \langle |z(t+\tau) - z(t)|^2 \rangle$$
  - If $z(t)$ is a random walk, $\mathrm{Var}(\tau) \sim \tau$.
  - If $z(t)$ has momentum, $\mathrm{Var}(\tau) \sim \tau^{2H}$, H>0.5.

Hurst Exponent and Variance Ratio Test

- Suppose we find H>0.5, what is the probability that the "real" H is actually 0.5?
- **Variance Ratio Test** will find probability that

$$\frac{Var\big(z(t)-z(t-\tau)\big)}{\tau Var\big(z(t)-z(t-1)\big)}$$

  is actually 1.
- Unfortunately, these tests only apply to long-term momentum, which is seldom present.

Hurst Exponent

- *Exercise*: Apply the hurst exponent test

Time Series Momentum Strategy

- Strategy:
  – Buy (sell) TU if it has a positive (negative) 12-month return, and hold for a month.
  – Start a new (staggered, “pyramided”) position every day within that month.

Indicators for TS Momentum

- Roll returns, sign(lagged-returns)
- Buy when the price reaches new N-day high
- Buy when the price exceeds MA or EMA
- Buy when the price exceeds upper Bollinger band
- Buy when num(up days) > num(down days)
- Alexander Filter: buy when daily return moves up at least x%, sell & short when price moves down at least x% from subsequent high.

Indicators for TS Momentum

- Best: combine long-term momentum indicator with short-term reversal indicator
- *Exercise*: Buy CL future at close if the price is lower than 30 days ago but is higher than 40 days ago. Vice versa for shorts.

Cross-sectional Stocks Momentum

- Cross-sectional momentum also present in S&P 500 stocks.
  - Total return = market return + factor returns + residual returns
  - Factor returns change slowly.
  - Long-short portfolio will hedge away market return.

- Strategy:
  - Rank the 1-year return of S&P 500 stocks every day.
  - (The universe can be arbitrarily large: constrained only by liquidity.)
  - Buy (short) the top (bottom) decile and hold for a month.

Momentum Crashes

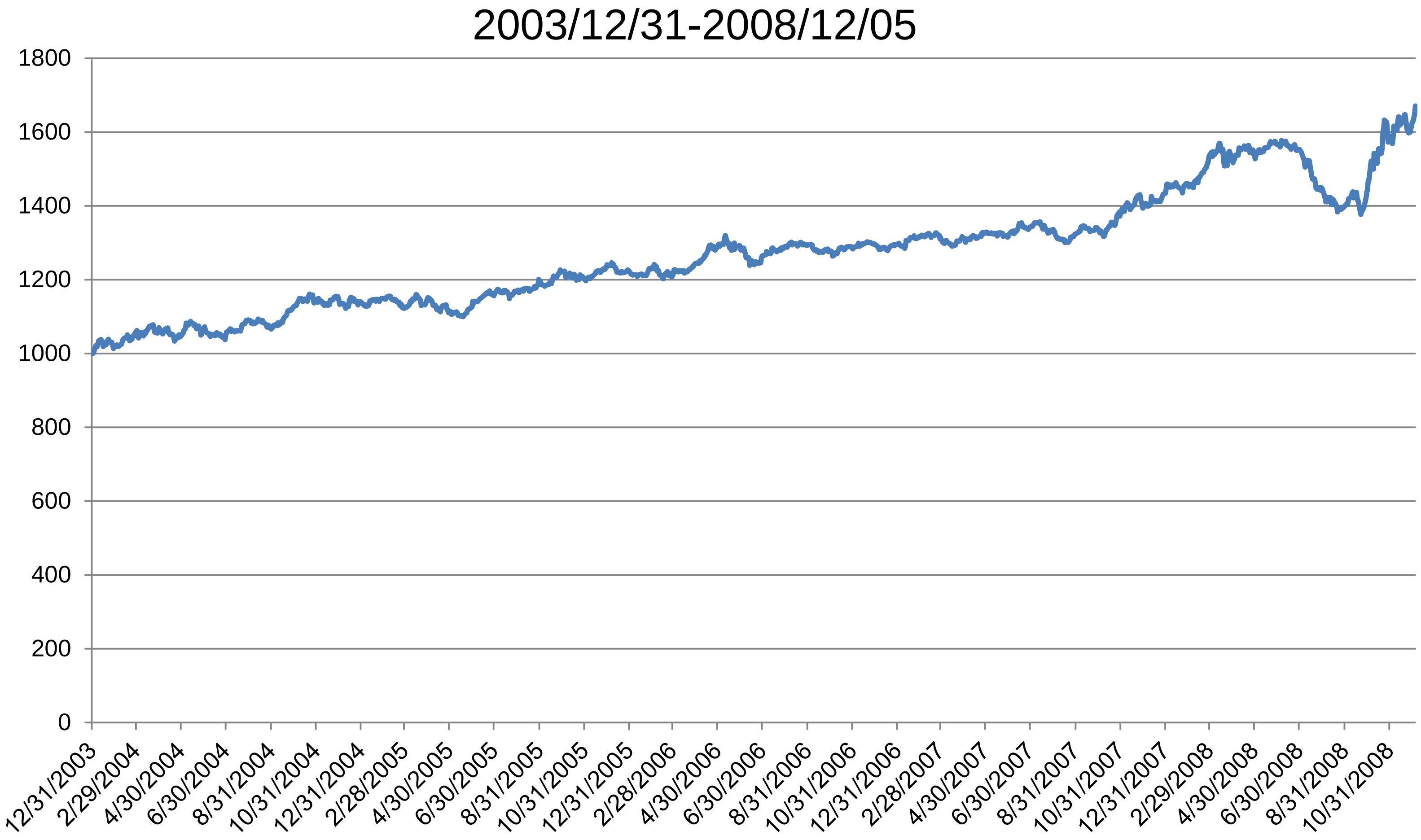
- Performance during 2008-2009 is very poor.
- This is common to most momentum strategies during and in the aftermath of a financial crisis.
  - Called “Momentum Crashes” by the researchers Kent Daniel *et. al*.
  - In the aftermath of the market crash of 1929, the drawdown duration for a prototypical momentum strategy is 30 years!
  - The main reason for “crash” is the strong rebound of short positions after a crash.

Momentum Crashes

- Look at a “momentum strategy index”: Diversified Trends Indicator SPDTP.
  - Mutual fund RYMFX and ETF WDTI are used to track this index.

	APR	Sharpe Ratio
2003/12/31-2008/12/05	10.8%	1.3
2009/01/02-2012/07/26	-8.2%	-1.0

DTI Index


2003/12/31-2008/12/05
1800
1600
1400
1200
1000
800
600
400
200
0
12/31/2003
2/29/2004
4/30/2004
6/30/2004
8/31/2004
10/31/2004
12/31/2004
2/28/2005
4/30/2005
6/30/2005
8/31/2005
10/31/2005
12/31/2005
2/28/2006
4/30/2006
6/30/2006
8/31/2006
10/31/2006
12/31/2006
2/28/2007
4/30/2007
6/30/2007
8/31/2007
10/31/2007
12/31/2007
2/29/2008
4/30/2008
6/30/2008
8/31/2008
10/31/2008

DTI Index


2008/12/08-2012/07/26
1800
1600
1400
1200
1000
800
600
400
200
0
12/8/2008
2/8/2009
4/8/2009
6/8/2009
8/8/2009
10/8/2009
12/8/2009
2/8/2010
4/8/2010
6/8/2010
8/8/2010
10/8/2010
12/8/2010
2/8/2011
4/8/2011
6/8/2011
8/8/2011
10/8/2011
12/8/2011
2/8/2012
4/8/2012
6/8/2012

Indicators for CS Momentum

- What "factors" can be used for ranking?
  - (Previously described) Lagged returns (or roll returns).
  - Fundamental macro-economic factors (futures)
  - Principal Component Analysis (futures/stocks)
  - Earnings growth (stocks)
  - Book-to-price ratio (stocks)
  - Linear combination of all of the above!

News Sentiment

- A new factor is available for stocks.
- Natural language processing algorithms are used to parse and analyze all news feed automatically.
- “Sentiment score” assigned to each story indicating possible price impact.
- Aggregation of sentiment score from fixed period is predictive of future returns.

News Sentiment

- Strategy: form a long-short portfolio using sentiment score as a ranking factor.
  - https://www.ravenpack.com/research/short-term-stock-selection-using-news/
  - The success of the strategy demonstrates the slow diffusion, analysis, and acceptance of news is cause of momentum.
- Instead of general sentiment, we can identify the predictive power of each specific type of event.

Event-driven Momentum

- E.g. Post Earnings Announcement Drift (PEAD)
- Studied since 1968, still profitable today!
  - See review by Bernard, Victor L., and Thomas, Jacob K. 1989. Post-Earnings-Announcement Drift: Delayed Price Response or Risk Premium? Journal of Accounting Research, 27, 1-36.
- The duration of momentum has decreased over the years.

PEAD

- Strategy:
  - Buy (sell) at open if there was an earnings announcement since the previous close, and the return since previous close retC2O is greater (less) than (-)0.5*stddev(retC2O, 90-day)
  - Exit at market close.
- *Exercise*: Backtest this strategy!
  - Stock prices in *inputDataOHLCDaily_stocks_20120424*, earnings announcements in *earnannfile* (earnann==1 if announced after previous MKT CLS and before today's MKT OPN.)
  - Try holding overnight too.

Other events driving momentum

- Earnings announcement is not the only event driving momentum.
- A partial list includes:
  - Earnings guidance
  - Analyst ratings change
  - Analyst recommendation change
  - Same-store sales announcement
  - Airline load factors announcement
  - Mergers and acquisitions announcement
  - Macroeconomic data release (ETF/Futures/FX)
  - Interest rate announcement (ETF/Futures/FX)

Other events driving momentum

- Contemporary, comprehensive study of events:
  Hafez, Peter A. 2011. Event Trading Using Market Response.
  https://www.ravenpack.com/research/event-trading-using-market-response/
  A surprise: After an M&A announcement, the acquiree's stock price falls more than the acquirer's.
- Some momentum duration may last only minutes.
  - E.g. BOE interest rate announcement induced GBPUSD momentum for 10 minutes. (Clare and Courtnenay, 2001).

Forced Sales and Purchases

- Next cause of momentum: Forced sales or purchases of assets by hedge/mutual/index/exchange-traded funds.
  - Contagion due to
    - Risk management (levered hedge funds)
    - Investor redemption/subscription (mutual funds)
  - Index composition changes
  - Levered ETFs: Forced rebalancing at market close.

Contagion leads to momentum

- Contagion due to risk management.
  - Suppose stock A is commonly long by **levered** hedge funds.
    - Suppose fund α suffers heavy, possibly unrelated loss.
    - The risk manager (via Kelly formula?) of α demands portfolio size reduction.
    - α sells stock A.
    - Stock A goes down in price.
    - Fund ß now suffers heavy loss if they hold A and other such stocks.
    - The risk manager of ß demands **deleveraging**.
    - ß sells stock A.
    - Stock A goes down further in price: contagion leads to momentum!
    - (Same logic for short positions.)
- Key driver: the need to maintain constant leverage in the face of loss.

Contagion leads to momentum

- This actually happened in August of 2007.
  - See Khandani, Amir, and Lo, Andrew. 2007. “What Happened to the Quants in August 2007?” https://web.mit.edu/Alo/www/Papers/august07.pdf
  - Unfortunately, hedge funds holdings change rapidly– hard to know which stock is commonly held, long or short, at any given time.
  - On the other hand, mutual funds holdings are more stable!

Mutual Funds Asset Fire Sale and Forced Purchases

- Contagion due to investors punishing/rewarding funds holding common stocks
  - Suppose stock A is commonly held by **mutual funds**.
    - Suppose fund α suffers heavy, possibly unrelated loss.
    - Investors of α redeem their investments.
    - α sells stock A.
    - Stock A goes down in price.
    - Fund ß now suffers heavy losses if they hold A and other such stocks.
    - Investors of ß redeem their investments.
    - ß sells stock A.
    - Stock A goes down further in price: contagion leads to momentum!
- Key driver: herding behavior of retail investors.

Mutual Funds Asset Fire Sale and Forced Purchases

- Note the condition that buying/selling of a stock is due to investor subscription/redemptions, not to investment decisions based on information specific to that stock.

Mutual Funds Asset Fire Sale and Forced Purchases

- Can form market-neutral portfolio based on the Pressure factor: long the top decile and short the bottom decile.
  - Re-balance quarterly.
  - Ann return=17% before transaction costs.
- If we predict the in/outflows of capital based on past performance, we can predict the future value of Pressure factor. (*i.e.* predict investor herding)
  - Another 17% ann return.
- Mean reversion of Pressure after 1 quarter
  - Another 7% ann return.

Mutual Funds Asset Fire Sale and Forced Purchases

- The general behavior of asset prices:
  - If price change is due to temporary liquidity demands (e.g. need to raise cash to satisfy investor redemptions) ⇒ mean reversion.
  - If price change is due to change in fundamentals ⇒price will move permanently to new level.
- Mutual fund holdings data from CRSP.
- The strategy has diminishing returns due to “crowding”.

Other Examples of Forced Sales and Purchases

- Index composition changes forced index funds to buy/sell stocks that are listed/delisted.
  - Momentum used to last multiple days.
    - Shankar, S. Gowri, and Miller, James M. 2006. Market Reaction to Changes in the S&P SmallCap 600 Index. The Financial Review, 41(3).
  - Recent tests suggest momentum reduced to intraday.

Levered ETF Momentum

- Levered ETFs must keep the ratio of the market value of holdings to net asset value constant at market close.
  - E.g. UPRO is levered 3x of SP 500 returns.
- If the market index return is negative (positive), the sponsor of the ETF needs to sell (buy) stocks to maintain leverage.
- *Exercise*: Suppose UPRO has about \$270M assets at the previous close. If SPX goes down 2%, what's the market value of the holdings it needs to sell?

Levered ETF Momentum

- Market value of holdings at previous close=3x$270M=$810M
- Decrease in NAV due to -2% market return=$16.2M
- NAV at today's close=
  $270M-$16.2M=$253.8M
- Target market value at today's close=
  3x$253.8M=$761.4M
- Market value of holdings at today's close=$810M-$16.2M=$793.8M
- Need to sell=$793.8-$761.4M=**$32.4M**

Levered ETF Momentum

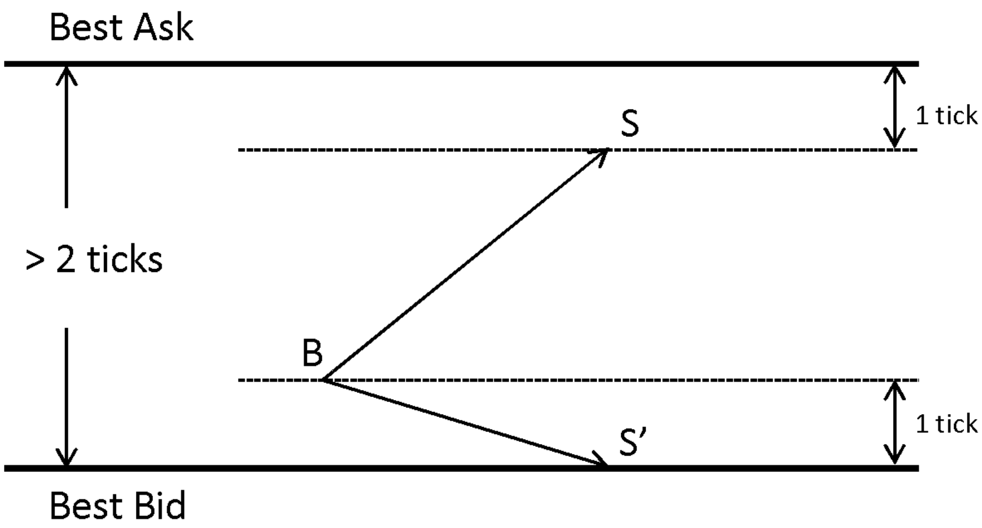
- This selling of $32.4M of holdings towards market close, in an already down market, generates momentum in UPRO, as well as in the SPX component stocks themselves.
- Strategy:
  - Buy(sell) UPRO if the return from the previous day's close to 15 minutes before today's close is greater(smaller) than (+/-)2%.
  - Exit at market close.
- APR=32%, Sharpe Ratio=2.2 from mid-2011 to mid-2012.

HFT: Ratio Trade

- Applies to futures markets which fill orders on a pro-rata basis (not time-priority). E.g. Eurodollars on CME.
  - Join the best bid if bid size ≫ ask size.
  - Once filled, place sell order at best ask.
  - If sell order not filled, sell at original best bid.

HFT: Ticking

- Applies if bid–ask spread $>$ 2 ticks.
- Place buy order at best bid + 1 tick if bid size $\gg$ ask size.
- Once filled, place sell order at best ask – 1 tick.
- If sell order not filled, sell at original best bid.


Best Ask
S
1 tick
> 2 ticks
B
S’
1 tick
Best Bid

HFT: Momentum Ignition

- If there is no natural momentum, you can ignite it!
- "Flipping"
  - Place large buy order at best bid, and a small sell order at best ask, to create impression of buying pressure.
  - Other traders are fooled into buying at ask due to the perception of buying momentum.
  - Once our sell order filled, cancel buy order.
  - Disappearance of buying "pressure" causes other traders to sell at bid.
  - We cover short position at bid.

HFT: Momentum Ignition

- “Stop Hunting”
  - Support and resistance levels are often well-known, often at round numbers.
    - E.g. $17.00 instead of $17.15.
  - Stop orders often cluster at these levels.
  - HFT can submit large sell orders near the support level so the price drops below support.
  - Sell-stop orders are triggered.
  - The price drops further.
  - HFT buy covers.
  - Ref. Osler, Carol. 2000. Support for Resistance: Technical Analysis and Intraday Exchange Rates. Federal Reserve Bank of New York Economic Policy Review 6 (July 2000): 53-65.

HFT: Order Flow

- Order flow leads price change
  - Order flow is *signed* transaction volume.
    - Buy *market* order is filled $\Rightarrow$ order flow > 0.
    - Sell *market* order is filled $\Rightarrow$ order flow < 0.
  - Order flow can be aggregated over some time interval to serve as indicator.
  - Researchers found that order flow is positively correlated with future price change.
    - Lyons, Richard. 2001. “The Microstructure Approach to Exchange Rates”
  - Therefore, order flow can be used to “front-run” market.

HFT: Order Flow

- How do we obtain order flow info if we are not large market makers or exchange operators?
  - In futures, can compute tick-by-tick whether trade executed at ask (positive flow) or bid (negative flow).
  - In stocks, more difficult due to fragmented markets and dark pools that do not report quotes and report only delayed trades.
  - In currencies, often impossible, but can monitor currency futures instead.
  - See also Easley *et al.* 2012. “Bulk Classification of Trading Activity”.
- Signed volume is a much better predictor of momentum than unsigned volume.
- (Note fund flow is a form of order flow.)

Exit Strategies

- Time-based
  - E.g. Gap strategies should exit at market close.
- New entry signal
  - E.g. if the return goes from positive to negative, flip position from long to short
- Stop loss
  - If the position incurs a loss, it means momentum w.r.t. entry price has reversed: exit is logical!
- Trailing stop loss
  - Momentum w.r.t. recent high has reversed: exit is logical!

Advantages of Momentum Strategies

- Ease of risk management
  - Stop loss is logical.
  - Loss ⇒ Momentum Reversed ⇒ Flatten or take opposite position.
  - "Black Swan" events usually generate abnormal upside, while downside is limited by stop loss.
    - High kurtosis in returns distribution benefits momentum strategies, but devastate mean-reverting ones.
- Diversification
  - Futures exhibit momentum due to roll returns.
  - Futures markets are more diversified than equities.
  - "Following The Trend: Diversified Managed Futures Trading" by Andreas Clenow

Disadvantages of Momentum Strategies

- Futures momentum often requires a long holding period.
  - Few independent trades.
  - Low Sharpe ratio and statistical significance.
- Event-driven momentum duration can shorten over time.
- Momentum crashes – underperformance for a prolonged period in the aftermath of a financial crisis.

Summary

- Causes of momentum:
  - Futures roll returns
    - Time-series and cross-sectional momentum strategies.
    - Arbitrage between futures and spot returns.
  - Slow diffusion of news
    - PEAD and other event-driven strategies.
  - Forced sales and purchases by funds.
    - Risk management contagion.
    - Investors redemptions/subscriptions contagion.
    - Index funds additions/deletions.
    - Levered ETFs rebalancing.
  - HFT momentum: manipulation of order book.